

Extreme Points of Positive-Definite Extension Sets

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Status

Candidate full solution, likely valid, for the locally compact Abelian measure-valued formulation of Question 3.1 in Jorgensen–Pedersen–Tian [1]. The result is an abstract characterization rather than a closed-form parametrization.

Source Problem

The source paper studies positive definite extensions of a continuous positive definite function F initially defined on a difference set. On page 65 it asks:

3.1 Locally Compact Abelian Groups

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Note that $\text{Ext}(F)$ may be empty. For $G = \mathbb{R}^2$, Rudin gave examples where $\text{Ext}(F) = \emptyset$ [Rud70, Rud63]. See Section 3.2.2, and the example of logarithmic Riemann surface in Section 4.6.

Question 3.1. Suppose $\text{Ext}(F) \neq \emptyset$, then what are its extreme points? Equivalently, characterize $\text{ext}(\text{Ext}(F))$.

The reader is referred to Section 3.2.3, especially, the discussion of $\text{Ext}_1(F)$ as a set of extreme points in $\text{Ext}(F)$; and the direct integral decomposition.

In the locally compact Abelian case, using the source’s Bochner-measure model, let G be the ambient group, let $\widehat{G}$ be its dual, and put

$$D = \Omega^{-1}\Omega.$$

After the source’s normalization $F(e) = 1$, write

$$\text{Ext}(F) = \left\{ \mu \in \mathcal{P}(\widehat{G}) : \widehat{\mu}(x) = F(x) \text{ for every } x \in D \right\},$$

where

$$\widehat{\mu}(x) = \int_{\widehat{G}} \langle \lambda, x \rangle d\mu(\lambda).$$

Proof Intuition

Convex decompositions of a measure μ inside $\text{Ext}(F)$ are exactly bounded signed perturbations of μ whose Fourier transform vanishes on the observed set $D = \Omega^{-1}\Omega$. Indeed, if $\mu = t\mu_1 + (1-t)\mu_2$, then μ_1 and μ_2 are automatically absolutely continuous with respect to μ , with bounded Radon–Nikodym derivatives. Conversely, any bounded real density h invisible to all observed characters produces the two distinct extension measures $(1 \pm \varepsilon h)\mu$. This gives the full characterization. In the Euclidean compact-support subcase, invisible perturbations vanish by analytic continuation, giving a concrete sufficient condition for extremality.

The Extremality Characterization

Theorem 1 (bounded perturbation characterization). *Let G be a locally compact Abelian group, let $\Omega \subset G$ be nonempty, let $D = \Omega^{-1}\Omega$, and let $F : D \rightarrow \mathbb{C}$ be continuous positive definite with $F(e) = 1$. Assume $\text{Ext}(F) \neq \emptyset$, where $\text{Ext}(F)$ is identified with the probability measures μ on $\widehat{G}$ satisfying $\widehat{\mu}|_D = F$. For $\mu \in \text{Ext}(F)$, the following are equivalent:*

1. μ is an extreme point of $\text{Ext}(F)$.
2. There is no nonzero real-valued $h \in L^\infty(\mu)$ such that

$$\int_{\widehat{G}} \langle \lambda, x \rangle h(\lambda) d\mu(\lambda) = 0, \quad x \in D.$$

3. The real linear span of

$$\{\text{Re}\langle \cdot, x \rangle, \text{Im}\langle \cdot, x \rangle : x \in D\}$$

is dense in $L^1(\mu; \mathbb{R})$.

Proof. We prove first the equivalence of (1) and (2). Suppose that μ is not extreme. Then

$$\mu = t\mu_1 + (1-t)\mu_2$$

for some $0 < t < 1$ and distinct $\mu_1, \mu_2 \in \text{Ext}(F)$. Since $\mu_i \ll \mu$, write $g_i = d\mu_i/d\mu$. The identity above gives $tg_1 + (1-t)g_2 = 1$, and in particular $g_i \in L^\infty(\mu)$. Since both μ_i and μ have the same Fourier transform on D ,

$$\int_{\widehat{G}} \langle \lambda, x \rangle (g_1(\lambda) - 1) d\mu(\lambda) = 0, \quad x \in D.$$

The real function $h = g_1 - 1$ is not zero, otherwise $\mu_1 = \mu$ and then also $\mu_2 = \mu$.

Conversely, let $h \in L^\infty(\mu)$ be real, nonzero, and annihilate all characters indexed by D . Since $0 \in D$, taking $x = 0$ gives $\int h d\mu = 0$. Choose $\varepsilon > 0$ so small that $1 \pm \varepsilon h \geq 0$ μ -almost everywhere, and define

$$d\mu_\pm = (1 \pm \varepsilon h) d\mu.$$

Then $\mu_\pm$ are probability measures, $\mu = (\mu_+ + \mu_-)/2$, and their Fourier transforms agree with $\widehat{\mu} = F$ on D . Hence $\mu_\pm \in \text{Ext}(F)$, and the decomposition is nontrivial.

Finally, (2) and (3) are the standard annihilator reformulation. The annihilator in $L^\infty(\mu; \mathbb{R})$ of the real span of the observed characters is precisely the set of real $h \in L^\infty(\mu)$ satisfying the displayed equations in (2). Thus (2) says that this annihilator is zero. By the Hahn–Banach separation theorem, a real linear subspace of $L^1(\mu; \mathbb{R})$ has zero annihilator in $L^\infty(\mu; \mathbb{R})$ if and only if it is norm dense. $\square$

Euclidean Compact-Support Consequences

Corollary 1. *Let $\Omega \subset \mathbb{R}^n$ be nonempty and open, and let $F : \Omega - \Omega \rightarrow \mathbb{C}$ be continuous and positive definite. If $\mu \in \text{Ext}(F)$ has compact support in $\mathbb{R}^n$, then μ is an extreme point of $\text{Ext}(F)$.*

Proof. By Lemma 1, it suffices to show that no nonzero bounded perturbation can have a Fourier transform vanishing on $D = \Omega - \Omega$. Let $h \in L^\infty(\mu)$ be real-valued and suppose

$$\int e^{ix \cdot \lambda} h(\lambda) d\mu(\lambda) = 0, \quad x \in D.$$

Set $d\sigma = h d\mu$. Since μ has compact support, σ is a compactly supported finite signed measure. Its Fourier transform extends to an entire function on $\mathbb{C}^n$:

$$\Phi(z) = \int_{\mathbb{R}^n} e^{iz \cdot \lambda} d\sigma(\lambda), \quad z \in \mathbb{C}^n.$$

The set D is a nonempty open subset of $\mathbb{R}^n$, and Φ vanishes on D . Therefore the real-analytic function $\Phi|_{\mathbb{R}^n}$ has all derivatives zero at every point of D , so the entire power series of Φ at any such point is zero. Hence $\Phi \equiv 0$ on $\mathbb{C}^n$.

In particular, the Fourier transform of σ vanishes on all of $\mathbb{R}^n$. By uniqueness of the Fourier transform for finite Borel measures on $\mathbb{R}^n$, $\sigma = 0$. Thus $h = 0$ in $L^\infty(\mu)$, and Lemma 1 implies that μ is extreme. $\square$

Corollary 2. *For a fixed F and $\Omega \subset \mathbb{R}^n$, the set $\text{Ext}(F)$ contains at most one compactly supported measure.*

Proof. If $\mu, \nu \in \text{Ext}(F)$ are compactly supported, then $\sigma = \mu - \nu$ is compactly supported and $\hat{\sigma} = 0$ on $\Omega - \Omega$. The same analytic-continuation and Fourier-uniqueness argument gives $\sigma = 0$. $\square$

Limitations And Novelty Check

The main theorem is a full abstract characterization of the extreme boundary of the measure-valued $\text{Ext}(F)$ in the locally compact Abelian setting of Question 3.1. It is not a parametrization by a small named family of measures, and it does not classify non-Abelian unitary-representation triples from the source's broader $\text{Ext}(F)$ formalism. The compact-support corollary uses Euclidean analytic continuation and is not asserted for arbitrary locally compact Abelian groups.

The cheap run indexes were searched for arXiv:1507.02547, the source title, positive-definite extension keywords, and extreme-point keywords. No exact prior packet was found before this work. Web searches for exact and close phrases around $\text{Ext}(F)$, positive definite extensions, bounded perturbations, L^∞ -criteria, Radon–Nikodym derivatives, compact support, and extreme points found the source paper and neighboring positive-definite-function literature, but no exact statement of the characterization above.

Human Review Recommendation

Review as a full abstract answer to Question 3.1 in the locally compact Abelian measure formulation. The main check is that this is the intended reading of the source question; under that reading, the theorem gives necessary and sufficient conditions for membership in $\text{ext}(\text{Ext}(F))$.

References

- [1] Palle Jorgensen, Steen Pedersen, and Feng Tian, *Extensions of Positive Definite Functions: Applications and Their Harmonic Analysis*, arXiv:1507.02547, 2015.
- [2] Walter Rudin, *Fourier Analysis on Groups*, Interscience Publishers, 1962.